CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

GONCALVES DAIRY

15782 ARROYO AVE

DOS PALOS CA 93620

NOTES:

PREPARED BY:

CARDOSO AG SERVICES
PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Goncalves Dairy

Physical address of dairy:

15782 Arroya AVE	Dos Palos	Merced	93620
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 12/01/1986

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X085-X017-X064-XXXX

B. OPERATORS

Goncalves, Frank

Operator name: Goncalves, Frank

Telephone no.: (209) 617-6324 (Landline) (Cellular)

15728 Arroya AVE	Dos Palos	CA	93620
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Goncalves, Frank

Legal owner name: Goncalves, Frank

Telephone no.: (209) 617-6324 (Landline) (Cellular)

15728 Arroya AVE	Dos Palos	CA	93620
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	245	148	120	97
Number under roof	640	110	0	0	0	0
Maximum number	640	110	245	148	120	97
Average number	640	110	245	148	120	97
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 23,029.13 *tons per reporting period*

Total nitrogen from manure: 285,066.77 *lbs per reporting period*

After ammonia losses (30% loss applied): 199,546.74 *lbs per reporting period*

Total phosphorus from manure: 46,444.57 *lbs per reporting period*

Total potassium from manure: 123,406.87 *lbs per reporting period*

Total salt from manure: 326,638.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 16,770,000 *gallons*

Total nitrogen generated: 101,630.94 *lbs*

Total phosphorus generated: 16,419.86 *lbs*

Total potassium generated: 375,431.87 *lbs*

Total salt generated: 1,188,568.17 *lbs*

	16,770,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	16,770,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
B Fields	78	78	2	process wastewater	X085-X180-X038-XXXX
Field 3	52	52	2	process wastewater	X085-X180-X023-XXXX
Field 4	56	56	2	manure	X085-X180-X023-XXXX
Field 5	55	55	2	manure	X085-X170-X064-XXXX
Field 6	44	44	2	both	X085-X170-X064-XXXX
Field 7	56	56	2	process wastewater	X085-X170-X064-XXXX
Field 8	26	26	2	process wastewater	X085-X170-X037-XXXX X085-X170-X064-XXXX
Field W3	19	19	2	manure	X085-X210-X060-XXXX
Totals for areas that were used for application	386	386	16		
Totals for areas that were not used for application					
Land application area totals	386	386	16		

B. CROPS AND HARVESTS

B Fields

Field name: B Fields

09/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 78 Plant date: 09/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	1,301.00 *ton*	Dry-weight		56.9	14,500.00	3,100.00	25,300.00		13.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.68	208.48	44.57	363.76	1,969.75

B Fields

05/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 78 Plant date: 05/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	2,412.00 *ton*	Dry-weight		69.6	7,000.00	1,400.00	67,000.00		4.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.92	131.61	26.32	1,259.68	752.05

Field 3

Field name: Field 3

09/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 52 Plant date: 09/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	892.00 *ton*	Dry-weight		65.4	14,200.00	2,300.00	15,700.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.15	168.56	27.30	186.37	1,163.31

05/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 52 Plant date: 05/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	1,734.00 *ton*	Dry-weight		74.7	9,300.00	1,700.00	8,300.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.35	156.92	28.68	140.05	809.91

Field 4

Field name: Field 4

09/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 56 Plant date: 09/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	876.00 *ton*	Dry-weight		54.7	12,200.00	1,700.00	12,100.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.64	172.90	24.09	171.49	1,162.14

05/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 56 Plant date: 05/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	1,765.00 *ton*	Dry-weight		69.3	10,600.00	1,900.00	8,000.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.52	205.13	36.77	154.82	1,006.30

Field 5

Field name: Field 5

09/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 09/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	693.00 *ton*	Dry-weight		49.5	12,900.00	1,800.00	11,800.00		8.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.60	164.17	22.91	150.17	1,119.89

Field 5

05/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 05/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	834.00 *ton*	Dry-weight		69.2	10,900.00	2,300.00	9,100.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	15.16	101.81	21.48	85.00	485.72

Field 6

Field name: Field 6

09/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 44 Plant date: 09/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	792.00 *ton*	Dry-weight		67.0	17,200.00	2,400.00	16,300.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	204.34	28.51	193.64	1,104.84

05/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 44 Plant date: 05/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	1,312.00 *ton*	Dry-weight		68.7	10,700.00	2,000.00	8,100.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.82	199.73	37.33	151.20	970.64

Field 7

Field name: Field 7

09/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 56 Plant date: 09/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	787.00 *ton*	Dry-weight		56.9	14,500.00	3,100.00	25,300.00		13.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.05	175.66	37.55	306.49	1,659.64

05/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 56 Plant date: 05/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	1,721.00 *ton*	Dry-weight		70.4	10,300.00	1,400.00	13,600.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	450.00	198.00	0.00
Total actual harvest content	30.73	187.39	25.47	247.43	1,109.80

Field 8

Field name: Field 8

09/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 09/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	366.00 *ton*	Dry-weight		56.9	14,500.00	3,100.00	25,300.00		13.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.08	175.95	37.62	307.00	1,662.40

Field 8

05/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 05/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	823.00 *ton*	Dry-weight		70.4	10,300.00	1,400.00	13,600.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.65	193.01	26.23	254.85	1,143.08

Field W3

Field name: Field W3

09/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 09/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	321.00 *ton*	Dry-weight		56.9	14,200.00	2,300.00	15,700.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.89	206.80	33.50	228.64	1,427.20

05/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 05/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	576.00 *ton*	Dry-weight		68.7	10,700.00	2,000.00	8,100.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.32	203.06	37.96	153.72	986.84

NUTRIENT BUDGET

A. LAND APPLICATIONS

B Fields - 09/18/2022: Oats, silage-soft dough

Field name: B Fields

Crop: Oats, silage-soft dough — Plant date: 09/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	107.40	15.12	186.26	605.85	936,000.00 *gal*
Canal	Surface water	0.20	0.00	0.00	0.00	18,791,394.00 *gal*
Application event totals		107.60	15.12	186.26	605.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	106.11	18.96	200.28	917.28	1,248,000.00 *gal*
Application event totals		106.11	18.96	200.28	917.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	61.44	10.60	322.45	1,023.43	1,092,000.00 *gal*
Canal	Surface water	0.18	0.00	0.00	0.00	16,646,972.00 *gal*
Application event totals		61.62	10.60	322.45	1,023.43	

B Fields - 05/10/2023: Corn, silage

Field name: B Fields

Crop: Corn, silage — Plant date: 05/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

B Fields - 05/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.20	0.00	0.00	0.00	18,791,394.00 *gal*
Application event totals		0.20	0.00	0.00	0.00	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	68.21	10.92	345.48	994.39	936,000.00 *gal*
Canal	Surface water	0.18	0.00	0.00	0.00	16,646,972.00 *gal*
Application event totals		68.38	10.92	345.48	994.39	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	56.84	9.10	287.90	828.66	780,000.00 *gal*
Canal	Surface water	0.21	0.00	0.00	0.00	19,840,792.00 *gal*
Application event totals		57.05	9.10	287.90	828.66	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	45.47	7.28	230.32	662.93	624,000.00 *gal*
Canal	Surface water	0.20	0.00	0.00	0.00	18,260,177.00 *gal*
Application event totals		45.67	7.28	230.32	662.93	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	13,769,275.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Field 3 - 09/18/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 09/18/2022

Field 3 - 09/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	93.98	13.23	162.98	530.12	546,000.00 *gal*
Canal	Surface water	0.15	0.00	0.00	0.00	9,170,826.00 *gal*
Application event totals		94.12	13.23	162.98	530.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	79.58	14.22	150.21	687.96	624,000.00 *gal*
Application event totals		79.58	14.22	150.21	687.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	52.66	9.08	276.39	877.23	624,000.00 *gal*
Canal	Surface water	0.15	0.00	0.00	0.00	9,086,092.00 *gal*
Application event totals		52.81	9.08	276.39	877.23	

Field 3 - 05/10/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	79.00	13.62	414.58	1,315.84	936,000.00 *gal*
Canal	Surface water	0.29	0.00	0.00	0.00	18,341,652.00 *gal*
Application event totals		79.29	13.62	414.58	1,315.84	

Field 3 - 05/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	18,172,184.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	59.68	9.55	302.30	870.09	546,000.00 *gal*
Canal	Surface water	0.21	0.00	0.00	0.00	12,895,863.00 *gal*
Application event totals		59.89	9.55	302.30	870.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	68.21	10.92	345.48	994.39	624,000.00 *gal*
Canal	Surface water	0.29	0.00	0.00	0.00	17,843,025.00 *gal*
Application event totals		68.49	10.92	345.48	994.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	14,212,499.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

Field 4 - 09/18/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 09/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 4 - 09/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	208.78	50.05	155.87	0.00	1,456.00 *ton*
Application event totals		208.78	50.05	155.87	0.00	

10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.13	0.00	0.00	0.00	8,940,523.00 *gal*
Application event totals		0.13	0.00	0.00	0.00	

Field 4 - 05/11/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 05/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	244.43	113.27	532.57	0.00	1,008.00 *ton*
Application event totals		244.43	113.27	532.57	0.00	

05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.20	0.00	0.00	0.00	13,410,785.00 *gal*
Application event totals		0.20	0.00	0.00	0.00	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.10	0.00	0.00	0.00	6,961,224.00 *gal*
Application event totals		0.10	0.00	0.00	0.00	

Field 4 - 05/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.24	0.00	0.00	0.00	16,115,755.00 *gal*
Application event totals		0.24	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.16	0.00	0.00	0.00	10,608,045.00 *gal*
Application event totals		0.16	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.28	0.00	0.00	0.00	18,706,660.00 *gal*
Application event totals		0.28	0.00	0.00	0.00	

Field 5 - 09/18/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 09/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	176.66	42.35	131.89	0.00	1,210.00 *ton*
Application event totals		176.66	42.35	131.89	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.13	0.00	0.00	0.00	8,888,379.00 *gal*
Application event totals		0.13	0.00	0.00	0.00	

Field 5 - 05/11/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 05/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	133.32	61.78	290.49	0.00	540.00 *ton*
Application event totals		133.32	61.78	290.49	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.20	0.00	0.00	0.00	13,332,569.00 *gal*
Application event totals		0.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.10	0.00	0.00	0.00	6,641,842.00 *gal*
Application event totals		0.10	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.26	0.00	0.00	0.00	16,907,692.00 *gal*
Application event totals		0.26	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	8,101,874.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.24	0.00	0.00	0.00	16,119,014.00 *gal*
Application event totals		0.24	0.00	0.00	0.00	

Field 6 - 09/06/2022: Oats, silage-soft dough

Field name: Field 6

Crop: Oats, silage-soft dough

Plant date: 09/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	208.78	50.05	155.87	0.00	1,144.00 *ton*
Application event totals		208.78	50.05	155.87	0.00	

10/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.16	0.00	0.00	0.00	8,365,853.00 *gal*
Application event totals		0.16	0.00	0.00	0.00	

Field 6 - 05/12/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 05/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	135.79	62.93	295.87	0.00	440.00 *ton*
Application event totals		135.79	62.93	295.87	0.00	

05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.32	0.00	0.00	0.00	16,731,706.00 *gal*
Application event totals		0.32	0.00	0.00	0.00	

Field 6 - 05/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	60.45	9.67	306.22	881.39	468,000.00 *gal*
Canal	Surface water	0.26	0.00	0.00	0.00	13,606,325.00 *gal*
Application event totals		60.71	9.67	306.22	881.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.38	0.00	0.00	0.00	19,804,943.00 *gal*
Application event totals		0.38	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	70.53	11.29	357.26	1,028.29	546,000.00 *gal*
Canal	Surface water	0.21	0.00	0.00	0.00	11,080,600.00 *gal*
Application event totals		70.74	11.29	357.26	1,028.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.41	0.00	0.00	0.00	21,522,436.00 *gal*
Application event totals		0.41	0.00	0.00	0.00	

Field 7 - 09/06/2022: Oats, silage-soft dough

Field name: Field 7

Crop: Oats, silage-soft dough

Plant date: 09/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 7 - 09/06/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	62.33	8.78	108.10	351.61	390,000.00 *gal*
Canal	Surface water	0.12	0.00	0.00	0.00	8,043,212.00 *gal*
Application event totals		62.45	8.78	108.10	351.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	62.33	8.78	108.10	351.61	390,000.00 *gal*
Application event totals		62.33	8.78	108.10	351.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	46.19	8.25	87.18	399.26	390,000.00 *gal*
Application event totals		46.19	8.25	87.18	399.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	46.19	8.25	87.18	399.26	390,000.00 *gal*
Application event totals		46.19	8.25	87.18	399.26	

Field 7 - 05/12/2023: Corn, silage

Field name: Field 7

Crop: Corn, silage

Plant date: 05/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	8,043,212.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Field 7 - 05/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	48.90	8.43	256.64	814.57	624,000.00 *gal*
Canal	Surface water	0.13	0.00	0.00	0.00	8,447,328.00 *gal*
Application event totals		49.03	8.43	256.64	814.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	48.90	8.43	256.64	814.57	624,000.00 *gal*
Canal	Surface water	0.20	0.00	0.00	0.00	13,211,986.00 *gal*
Application event totals		49.10	8.43	256.64	814.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	79.17	12.67	401.01	1,154.20	780,000.00 *gal*
Canal	Surface water	0.19	0.00	0.00	0.00	12,501,524.00 *gal*
Application event totals		79.35	12.67	401.01	1,154.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	71.25	11.40	360.91	1,038.78	702,000.00 *gal*
Canal	Surface water	0.19	0.00	0.00	0.00	12,501,524.00 *gal*
Application event totals		71.44	11.40	360.91	1,038.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.20	0.00	0.00	0.00	13,238,058.00 *gal*
Application event totals		0.20	0.00	0.00	0.00	

Field 8 - 09/06/2022: Oats, silage-soft dough

Field 8 - 09/06/2022: Oats, silage-soft dough

Field name: Field 8

Crop: Oats, silage-soft dough

Plant date: 09/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	107.40	15.12	186.26	605.85	312,000.00 *gal*
Canal	Surface water	0.12	0.00	0.00	0.00	3,876,037.00 *gal*
Application event totals		107.52	15.12	186.26	605.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	119.37	21.33	225.32	1,031.94	468,000.00 *gal*
Application event totals		119.37	21.33	225.32	1,031.94	

Field 8 - 05/12/2023: Corn, silage

Field name: Field 8

Crop: Corn, silage

Plant date: 05/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.19	0.00	0.00	0.00	5,814,056.00 *gal*
Application event totals		0.19	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	85.26	13.64	431.85	1,242.99	390,000.00 *gal*
Canal	Surface water	0.17	0.00	0.00	0.00	5,388,757.00 *gal*
Application event totals		85.43	13.64	431.85	1,242.99	

Field 8 - 05/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	102.31	16.37	518.22	1,491.59	468,000.00 *gal*
Canal	Surface water	0.17	0.00	0.00	0.00	5,388,757.00 *gal*
Application event totals		102.48	16.37	518.22	1,491.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	68.21	10.92	345.48	994.39	312,000.00 *gal*
Canal	Surface water	0.18	0.00	0.00	0.00	5,572,890.00 *gal*
Application event totals		68.38	10.92	345.48	994.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	5,572,890.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

Field W3 - 09/18/2022: Oats, silage-soft dough

Field name: Field W3

Crop: Oats, silage-soft dough

Plant date: 09/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	240.90	57.75	179.85	0.00	570.00 *ton*
Application event totals		240.90	57.75	179.85	0.00	

Field W3 - 09/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.14	0.00	0.00	0.00	3,198,165.00 *gal*
Application event totals		0.14	0.00	0.00	0.00	

Field W3 - 05/10/2023: Corn, silage

Field name: Field W3

Crop: Corn, silage

Plant date: 05/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	244.43	113.27	532.57	0.00	342.00 *ton*
Application event totals		244.43	113.27	532.57	0.00	

05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.21	0.00	0.00	0.00	4,797,248.00 *gal*
Application event totals		0.21	0.00	0.00	0.00	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.26	0.00	0.00	0.00	5,980,265.00 *gal*
Application event totals		0.26	0.00	0.00	0.00	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.26	0.00	0.00	0.00	5,980,265.00 *gal*
Application event totals		0.26	0.00	0.00	0.00	

Field W3 - 05/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.20	0.00	0.00	0.00	4,474,607.00 *gal*
Application event totals		0.20	0.00	0.00	0.00	

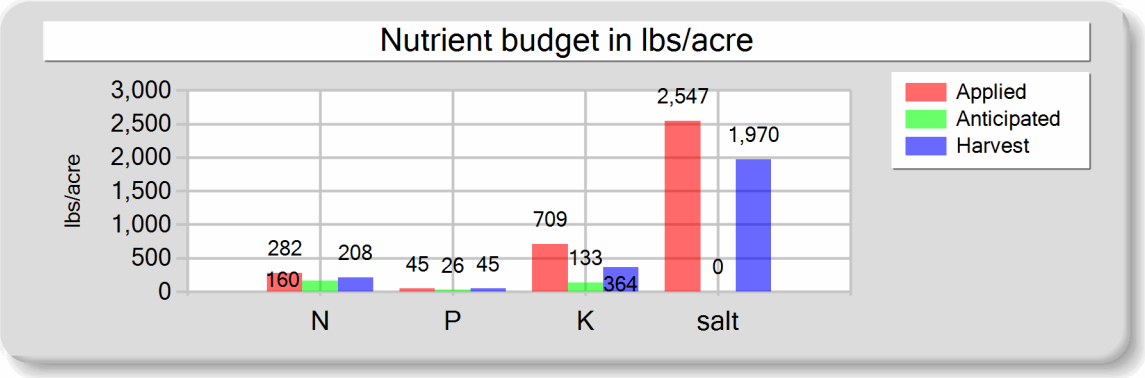
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.19	0.00	0.00	0.00	4,308,398.00 *gal*
Application event totals		0.19	0.00	0.00	0.00	

B. NUTRIENT BUDGET

B Fields - 09/18/2022: Oats, silage-soft dough

Field name: B Fields Crop: Oats, silage-soft dough Plant date: 09/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	274.95	44.68	708.99	2,546.56
Fresh water	0.38	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.33	44.68	708.99	2,546.56
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	208.48	44.57	363.76	1,969.75
Nutrient balance	73.85	0.11	345.23	576.81
Applied to removed ratio	1.35	1.00	1.95	1.29

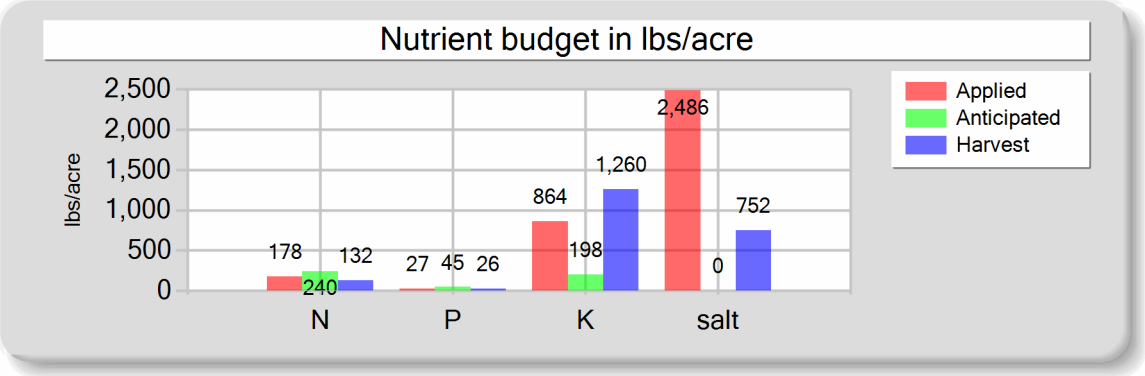
Fresh water applied	
35,438,366.00	*gallons*
1,305.07	*acre-inches*
16.73	*inches/acre*

Process wastewater applied	
3,276,000.00	*gallons*
120.64	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

B Fields - 05/10/2023: Corn, silage

Field name: B Fields Crop: Corn, silage Plant date: 05/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	170.51	27.29	863.71	2,485.98
Fresh water	0.93	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	178.45	27.29	863.71	2,485.98
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	131.61	26.32	1,259.68	752.05
Nutrient balance	46.84	0.97	-395.97	1,733.93
Applied to removed ratio	1.36	1.04	0.69	3.31

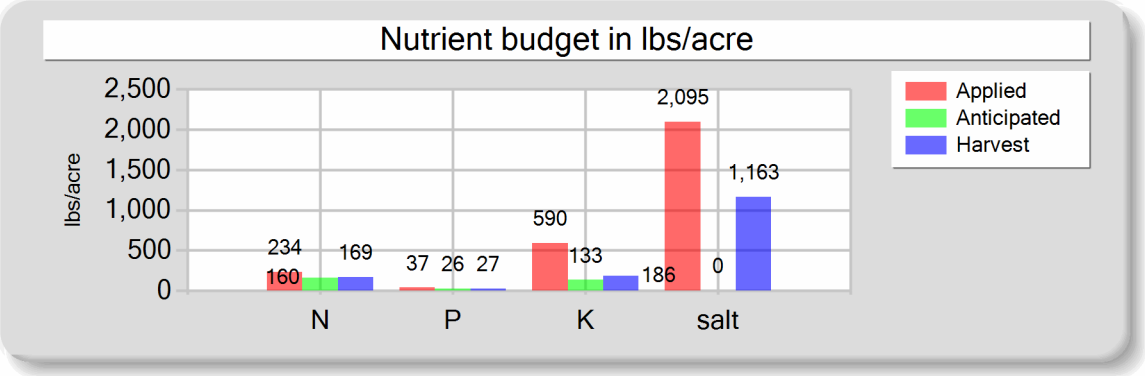
Fresh water applied	
87,308,610.00	*gallons*
3,215.28	*acre-inches*
41.22	*inches/acre*

Process wastewater applied	
2,340,000.00	*gallons*
86.17	*acre-inches*
1.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 09/18/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 09/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	226.22	36.53	589.57	2,095.30
Fresh water	0.29	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	233.51	36.53	589.57	2,095.30
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	168.56	27.30	186.37	1,163.31
Nutrient balance	64.95	9.23	403.21	932.00
Applied to removed ratio	1.39	1.34	3.16	1.80

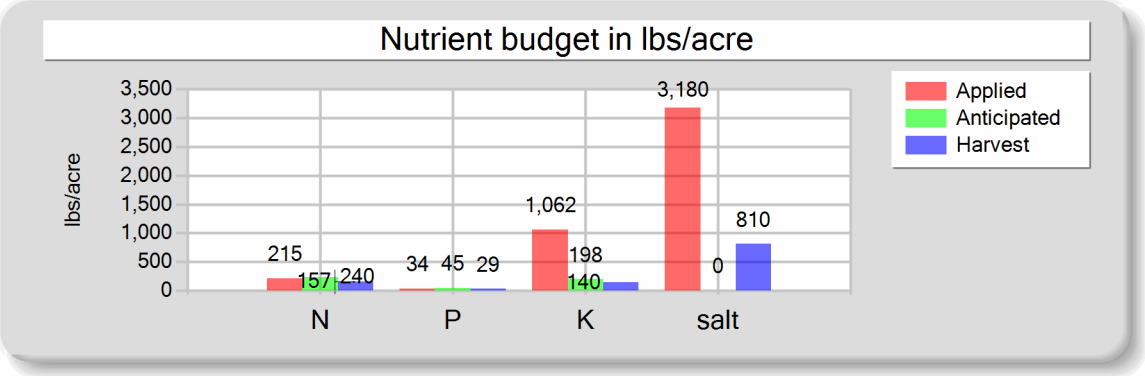
Fresh water applied	
18,256,918.00	*gallons*
672.34	*acre-inches*
12.93	*inches/acre*

Process wastewater applied	
1,794,000.00	*gallons*
66.07	*acre-inches*
1.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/10/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	206.88	34.09	1,062.36	3,180.32
Fresh water	1.31	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.19	34.09	1,062.36	3,180.32
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	156.92	28.68	140.05	809.91
Nutrient balance	58.27	5.41	922.31	2,370.41
Applied to removed ratio	1.37	1.19	7.59	3.93

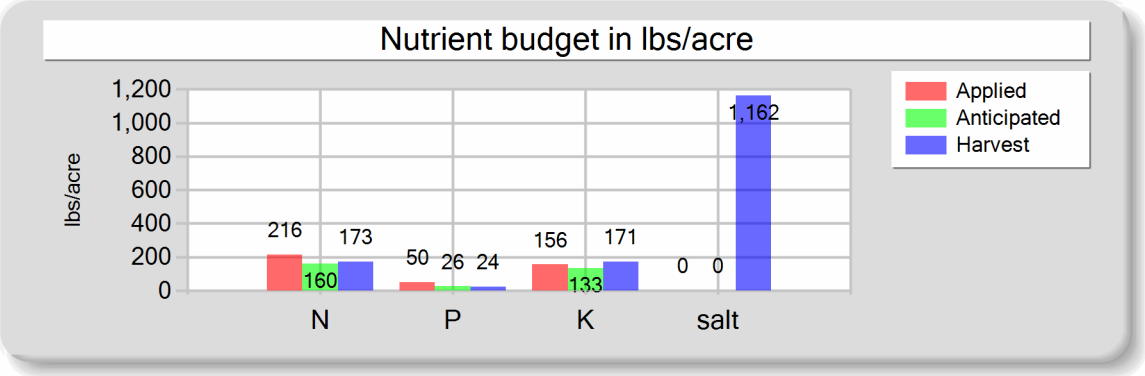
Fresh water applied	
81,465,223.00	*gallons*
3,000.09	*acre-inches*
57.69	*inches/acre*

Process wastewater applied	
2,106,000.00	*gallons*
77.56	*acre-inches*
1.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 09/18/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 09/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	208.78	50.05	155.87	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.13	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.91	50.05	155.87	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	172.90	24.09	171.49	1,162.14
Nutrient balance	43.01	25.96	-15.62	-1,162.14
Applied to removed ratio	1.25	2.08	0.91	0.00

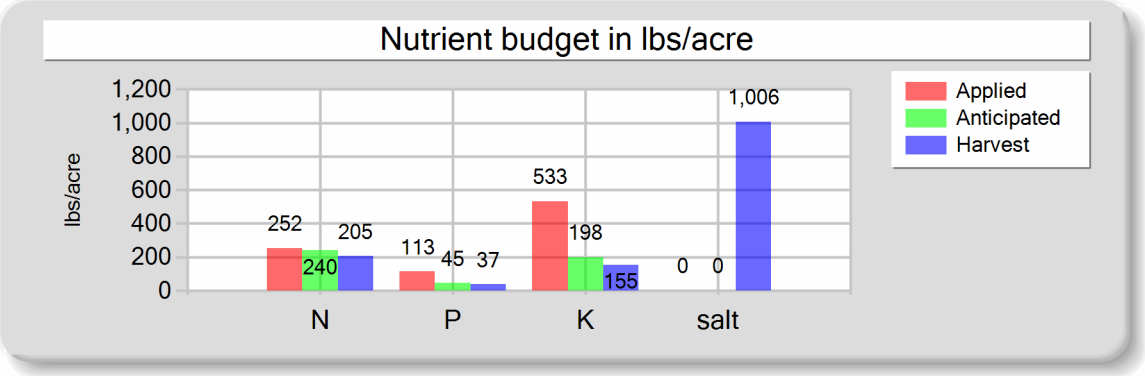
Fresh water applied	
8,940,523.00	*gallons*
329.25	*acre-inches*
5.88	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 05/11/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 05/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	244.43	113.27	532.57	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.98	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	252.41	113.27	532.57	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	205.13	36.77	154.82	1,006.30
Nutrient balance	47.28	76.50	377.75	-1,006.30
Applied to removed ratio	1.23	3.08	3.44	0.00

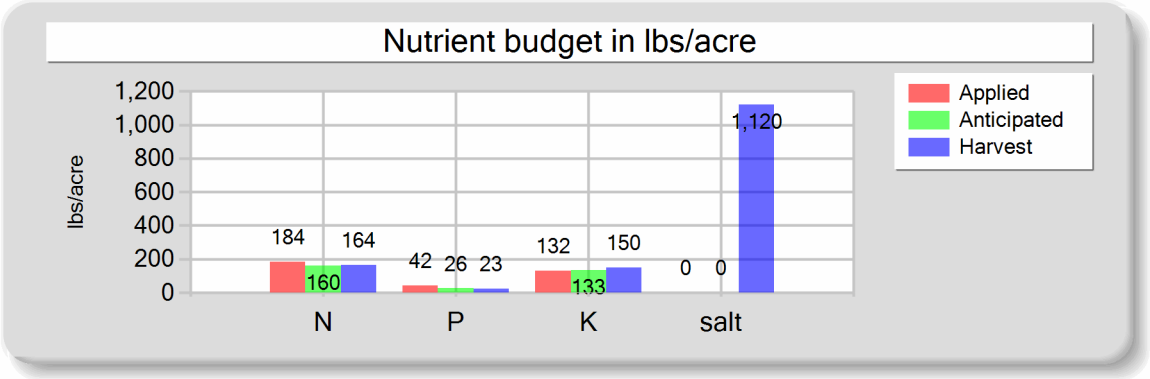
Fresh water applied	
65,802,469.00	*gallons*
2,423.28	*acre-inches*
43.27	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 09/18/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 09/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	176.66	42.35	131.89	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.13	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.79	42.35	131.89	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	164.17	22.91	150.17	1,119.89
Nutrient balance	19.63	19.44	-18.28	-1,119.89
Applied to removed ratio	1.12	1.85	0.88	0.00

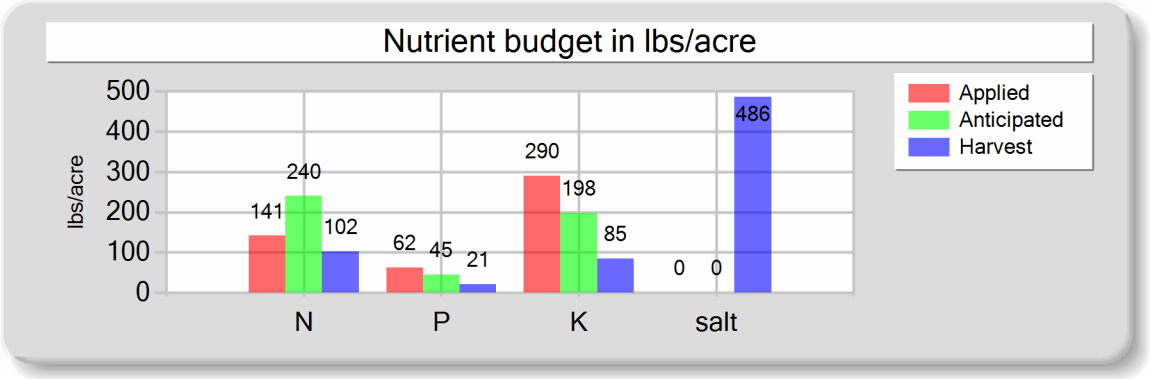
Fresh water applied	
8,888,379.00	*gallons*
327.33	*acre-inches*
5.95	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 05/11/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 05/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	133.32	61.78	290.49	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.93	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	141.25	61.78	290.49	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	101.81	21.48	85.00	485.72
Nutrient balance	39.44	40.30	205.49	-485.72
Applied to removed ratio	1.39	2.88	3.42	0.00

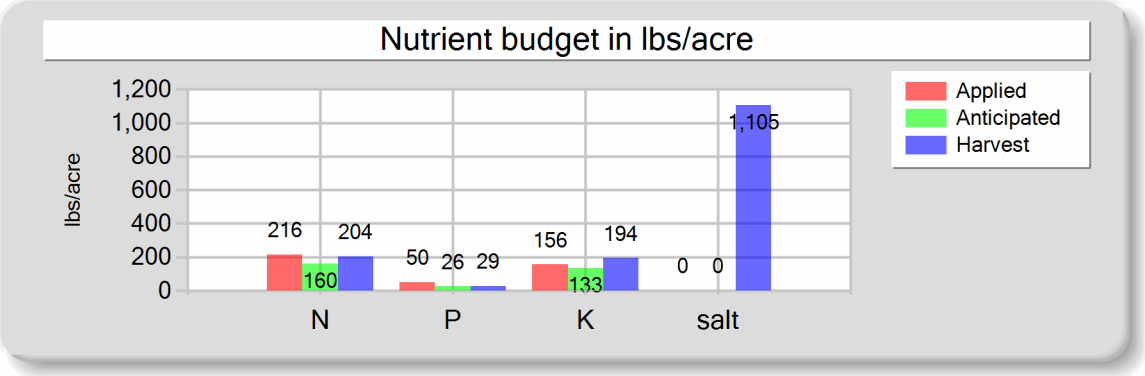
Fresh water applied	
61,102,991.00	*gallons*
2,250.22	*acre-inches*
40.91	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 09/06/2022: Oats, silage-soft dough

Field name: Field 6 Crop: Oats, silage-soft dough Plant date: 09/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	208.78	50.05	155.87	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.16	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.94	50.05	155.87	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	204.34	28.51	193.64	1,104.84
Nutrient balance	11.60	21.54	-37.77	-1,104.84
Applied to removed ratio	1.06	1.76	0.80	0.00

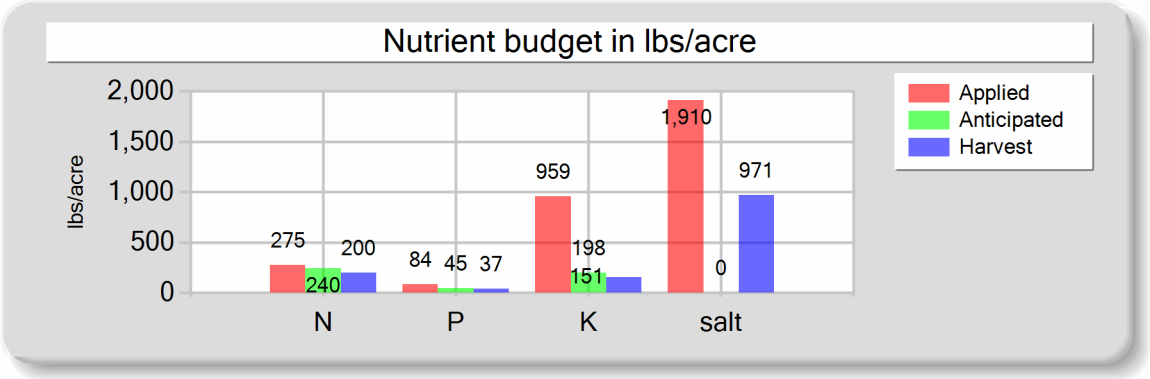
Fresh water applied	
8,365,853.00	*gallons*
308.09	*acre-inches*
7.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 05/12/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 05/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	135.79	62.93	295.87	0.00
Process wastewater	130.99	20.96	663.48	1,909.68
Fresh water	1.57	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	275.35	83.89	959.36	1,909.68
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	199.73	37.33	151.20	970.64
Nutrient balance	75.62	46.56	808.16	939.04
Applied to removed ratio	1.38	2.25	6.35	1.97

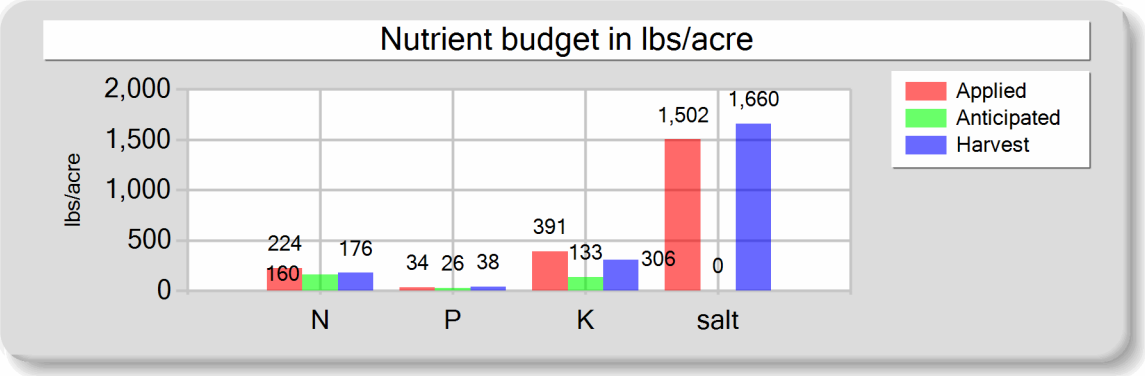
Fresh water applied	
82,746,010.00	*gallons*
3,047.25	*acre-inches*
69.26	*inches/acre*

Process wastewater applied	
1,014,000.00	*gallons*
37.34	*acre-inches*
0.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 09/06/2022: Oats, silage-soft dough

Field name: Field 7 Crop: Oats, silage-soft dough Plant date: 09/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	217.03	34.06	390.55	1,501.74
Fresh water	0.12	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.15	34.06	390.55	1,501.74
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	175.66	37.55	306.49	1,659.64
Nutrient balance	48.50	-3.50	84.06	-157.90
Applied to removed ratio	1.28	0.91	1.27	0.90

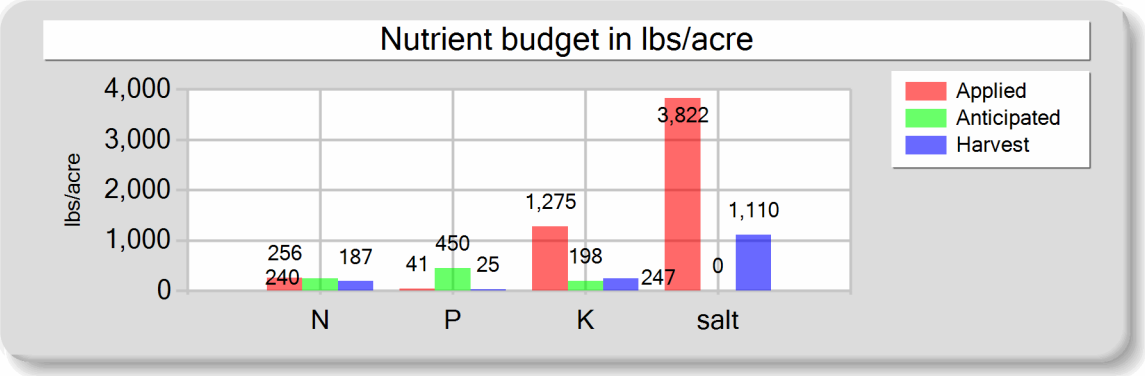
Fresh water applied	
8,043,212.00	*gallons*
296.20	*acre-inches*
5.29	*inches/acre*

Process wastewater applied	
1,560,000.00	*gallons*
57.45	*acre-inches*
1.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 05/12/2023: Corn, silage

Field name: Field 7 Crop: Corn, silage Plant date: 05/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	248.22	40.94	1,275.20	3,822.12
Fresh water	1.01	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	256.23	40.94	1,275.20	3,822.12
Anticipated crop nutrient removal	240.00	450.00	198.00	0.00
Actual crop nutrient removal	187.39	25.47	247.43	1,109.80
Nutrient balance	68.84	15.47	1,027.77	2,712.32
Applied to removed ratio	1.37	1.61	5.15	3.44

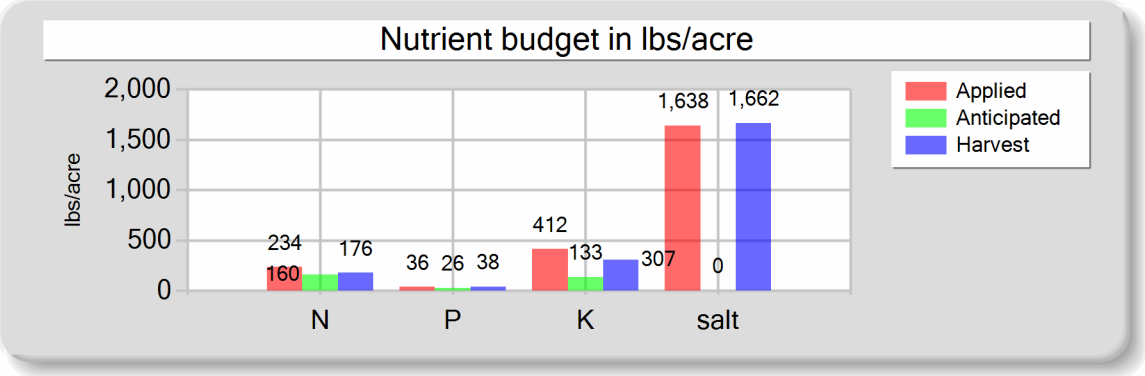
Fresh water applied	
67,943,632.00	*gallons*
2,502.13	*acre-inches*
44.68	*inches/acre*

Process wastewater applied	
2,730,000.00	*gallons*
100.54	*acre-inches*
1.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 09/06/2022: Oats, silage-soft dough

Field name: Field 8 Crop: Oats, silage-soft dough Plant date: 09/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	226.77	36.45	411.58	1,637.79
Fresh water	0.12	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	233.90	36.45	411.58	1,637.79
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	175.95	37.62	307.00	1,662.40
Nutrient balance	57.95	-1.17	104.58	-24.61
Applied to removed ratio	1.33	0.97	1.34	0.99

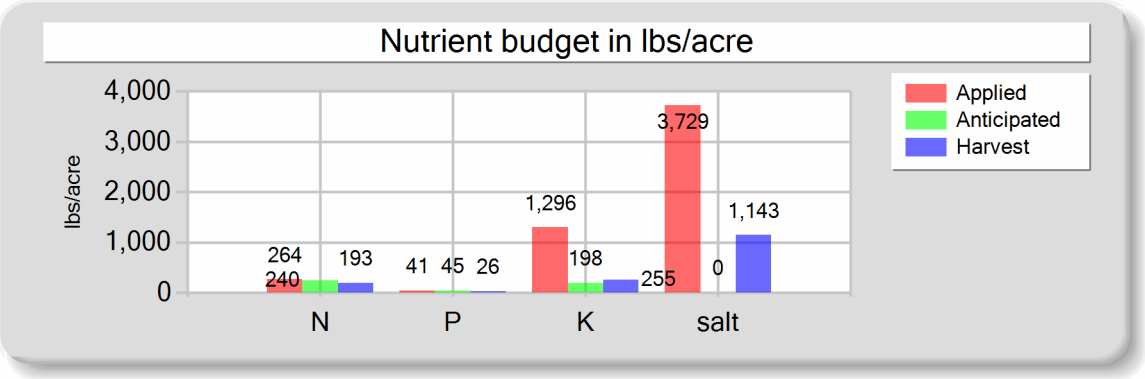
Fresh water applied	
3,876,037.00	*gallons*
142.74	*acre-inches*
5.49	*inches/acre*

Process wastewater applied	
780,000.00	*gallons*
28.72	*acre-inches*
1.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 05/12/2023: Corn, silage

Field name: Field 8 Crop: Corn, silage Plant date: 05/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	255.77	40.93	1,295.56	3,728.96
Fresh water	0.89	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	263.66	40.93	1,295.56	3,728.96
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	193.01	26.23	254.85	1,143.08
Nutrient balance	70.65	14.70	1,040.71	2,585.88
Applied to removed ratio	1.37	1.56	5.08	3.26

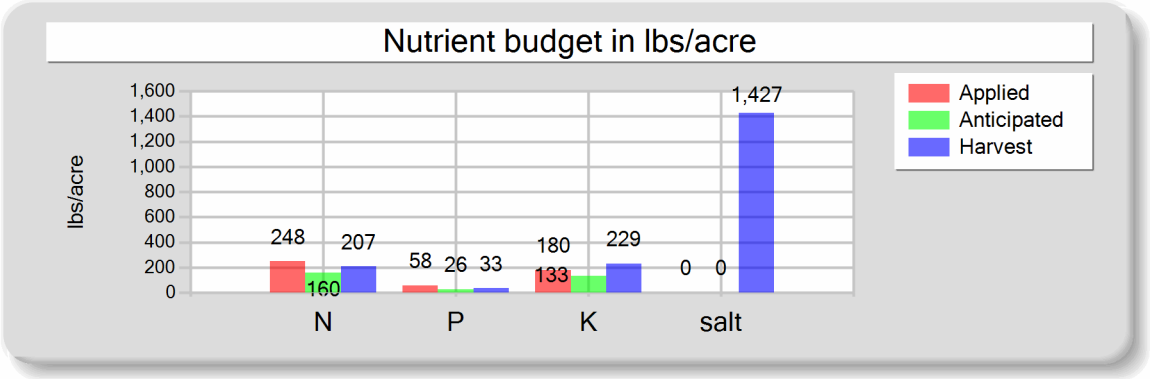
Fresh water applied	
27,737,350.00	*gallons*
1,021.47	*acre-inches*
39.29	*inches/acre*

Process wastewater applied	
1,170,000.00	*gallons*
43.09	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field W3 - 09/18/2022: Oats, silage-soft dough

Field name: Field W3 Crop: Oats, silage-soft dough Plant date: 09/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	240.90	57.75	179.85	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.14	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	248.04	57.75	179.85	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	206.80	33.50	228.64	1,427.20
Nutrient balance	41.24	24.25	-48.79	-1,427.20
Applied to removed ratio	1.20	1.72	0.79	0.00

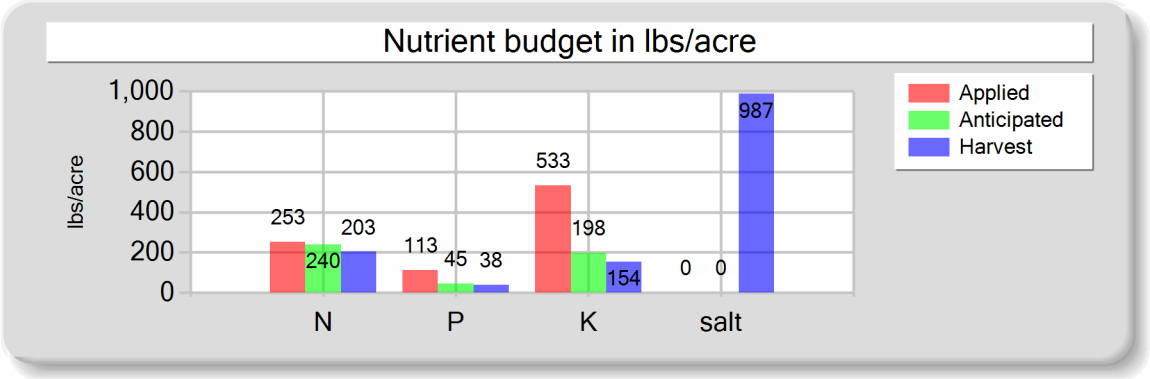
Fresh water applied	
3,198,165.00	*gallons*
117.78	*acre-inches*
6.20	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field W3 - 05/10/2023: Corn, silage

Field name: Field W3 Crop: Corn, silage Plant date: 05/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	244.43	113.27	532.57	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.12	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	252.55	113.27	532.57	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	203.06	37.96	153.72	986.84
Nutrient balance	49.49	75.32	378.85	-986.84
Applied to removed ratio	1.24	2.98	3.46	0.00

Fresh water applied	
25,540,783.00	*gallons*
940.58	*acre-inches*
49.50	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall 2022 Manure

Sample and source description: Fall 2022 Manure

Sample date: 10/10/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	3,500.00	10,900.00							
DL	1.00	1.00	1.00							

Corral Manure

Sample and source description: Corral Manure

Sample date: 05/15/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 44.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	5,700.00	26,800.00							0.00
DL	0.01	0.01	0.01							1.00

Corral Manure Spread

Sample and source description: Corral Manure Spread

Sample date: 09/26/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 40.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	4,700.00	16,300.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

2022 Q4 Lagoon

Sample and source description: 2022 Q4 Lagoon

Sample date: 12/12/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,070.00	510.00	0.00	2.50	151.00	1,860.00								15,200.00	6,050
DL	1.00	1.00	0.01	1.00	1.00	1.00								1.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	793.00	404.00	0.00	1.70	142.00	1,500.00								12,400.00	6,870
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	525.00	242.00	0.00	0.90	90.70	2,760.00								13,800.00	8,760
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 08/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	680.00	203.00	0.00	1.10	109.00	3,450.00								20,000.00	9,930
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	679.00	381.00	0.00	1.30	104.00	2,530.00								16,500.00	10,000
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

San Luis Canal Company

Sample description: San Luis Canal Company

Sample date: 08/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.10	0.00	0.10								213.00	
DL	0.01	0.01	0.01								1.00	

Domestic Well

DW - East

Sample description: DW - East

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.20	0.00	0.20								1,080.00	0
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW West

Sample description: DW West

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.20	0.00	0.20								1,080.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

B Fields - 09/18/2022: Oats, silage-soft dough

B Fields Oat Silage

Sample and source description: B Fields Oat Silage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	3,100.00	25,300.00		13.70
DL	0.01	0.01	0.01		1.00

B Fields - 05/10/2023: Corn, silage

B Fields - 05/10/2023: Corn, silage

Corn Fields B1, B2, B3, B4, B5

Sample and source description: Corn Fields B1, B2, B3, B4, B5

Sample date: 09/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,000.00	1,400.00	67,000.00		4.00
DL	0.01	0.01	0.01		1.00

Field 3 - 09/18/2022: Oats, silage-soft dough

Field 3 Winter Forage

Sample and source description: Field 3 Winter Forage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	2,300.00	15,700.00		9.80
DL	0.01	0.01	0.01		1.00

Field 3 - 05/10/2023: Corn, silage

Corn Field 3

Sample and source description: Corn Field 3

Sample date: 09/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,300.00	1,700.00	8,300.00		4.80
DL	0.01	0.01	0.01		1.00

Field 4 - 09/18/2022: Oats, silage-soft dough

Field 4 Winter Forage

Sample and source description: Field 4 Winter Forage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,200.00	1,700.00	12,100.00		8.20
DL	0.01	0.01	0.01		1.00

Field 4 - 05/11/2023: Corn, silage

Corn Field 4

Sample and source description: Corn Field 4

Sample date: 09/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	1,900.00	8,000.00		5.20
DL	0.01	0.01	0.01		1.00

Field 5 - 09/18/2022: Oats, silage-soft dough

Field 5 Winter Forage

Sample and source description: Field 5 Winter Forage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	1,800.00	11,800.00		8.80
DL	0.01	0.01	0.01		1.00

Field 5 - 05/11/2023: Corn, silage

Corn Field 5

Sample and source description: Corn Field 5

Sample date: 09/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	2,300.00	9,100.00		5.20
DL	0.01	0.01	0.01		1.00

Field 6 - 09/06/2022: Oats, silage-soft dough

Field 6 W3 Winter Forage

Sample and source description: Field 6 W3 Winter Forage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	2,400.00	16,300.00		9.30
DL	0.01	0.01	0.01		1.00

Field 6 - 05/12/2023: Corn, silage

Field 6 Corn Silage

Sample and source description: Field 6 Corn Silage

Sample date: 09/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	2,000.00	8,100.00		5.20
DL	0.01	0.01	0.01		1.00

Field 7 - 09/06/2022: Oats, silage-soft dough

Field 7

Sample and source description: Field 7

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	3,100.00	25,300.00		13.70
DL	1.00	1.00	1.00		1.00

Field 7 - 05/12/2023: Corn, silage

Field 7 Corn Silage

Sample and source description: Field 7 Corn Silage

Sample date: 09/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,300.00	1,400.00	13,600.00		6.10
DL	0.01	0.01	0.01		1.00

Field 8 - 09/06/2022: Oats, silage-soft dough

Field 8 Oat Silage

Sample and source description: Field 8 Oat Silage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	3,100.00	25,300.00		13.70
DL	0.01	0.01	0.01		1.00

Field 8 - 05/12/2023: Corn, silage

Field 8 Corn Silage

Sample and source description: Field 8 Corn Silage

Sample date: 09/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,300.00	1,400.00	13,600.00		6.10
DL	0.01	0.01	0.01		1.00

Field W3 - 09/18/2022: Oats, silage-soft dough

Oats

Sample and source description: Oats

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	2,300.00	15,700.00		9.80
DL	1.00	1.00	1.00		1.00

Field W3 - 05/10/2023: Corn, silage

Field W3 Corn Silage

Sample and source description: Field W3 Corn Silage

Sample date: 09/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	2,000.00	8,100.00		5.20
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

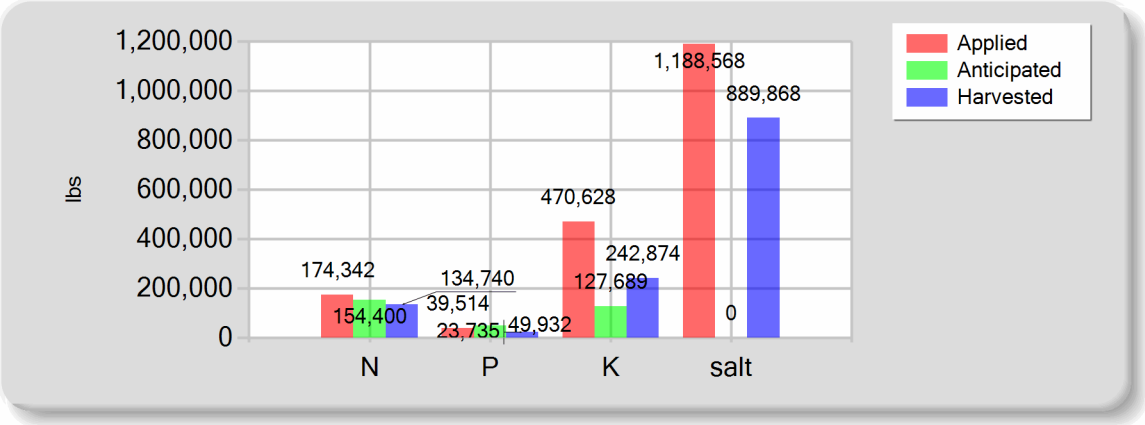
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

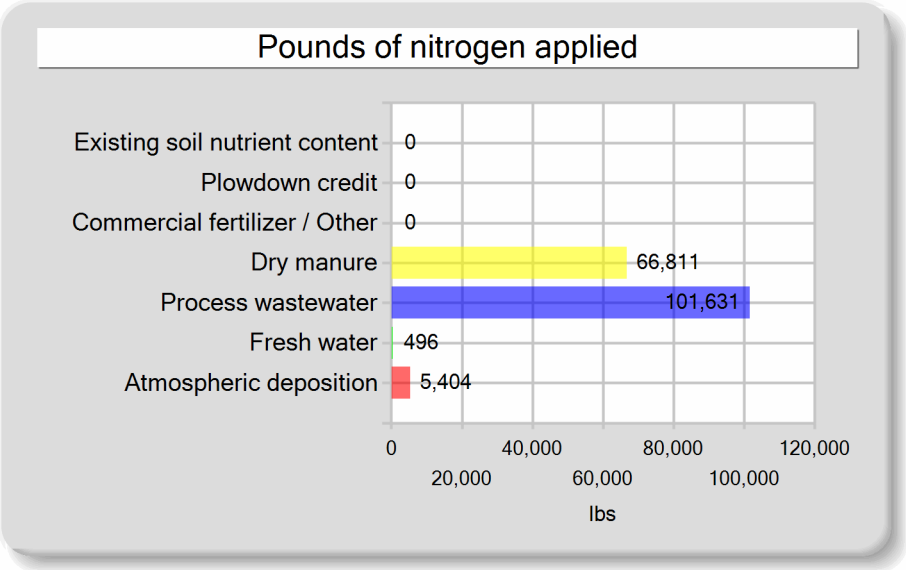
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

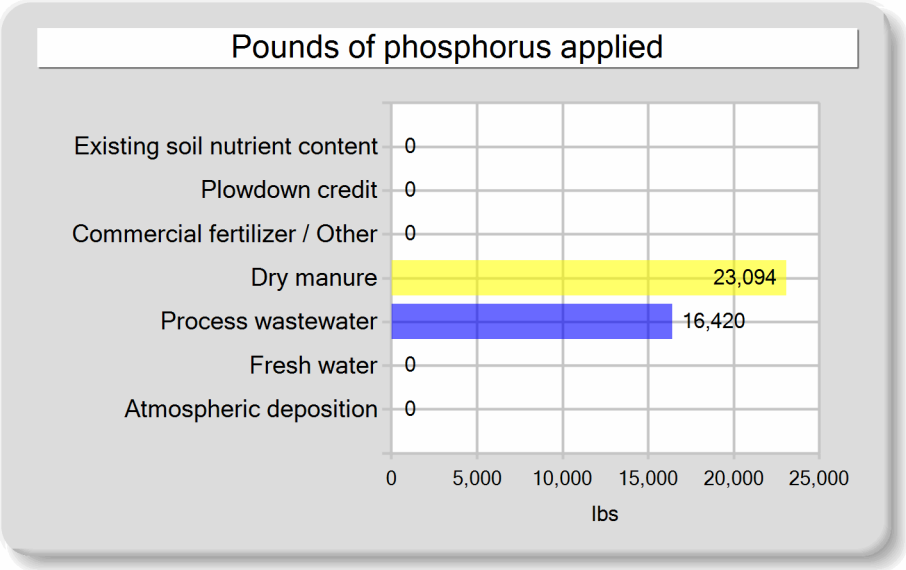
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	66,810.94	23,093.72	95,196.28	0.00
Process wastewater	101,630.94	16,419.86	375,431.87	1,188,568.17
Fresh water	496.24	0.00	0.00	0.00
Atmospheric deposition	5,404.00	0.00	0.00	0.00
Total nutrients applied	174,342.11	39,513.58	470,628.15	1,188,568.17
Anticipated crop nutrient removal	154,400.00	49,931.60	127,688.80	0.00
Actual crop nutrient removal	134,740.31	23,734.90	242,874.30	889,867.88
Nutrient balance	39,601.81	15,778.68	227,753.85	298,700.29
Applied to removed ratio	1.29	1.66	1.94	1.34

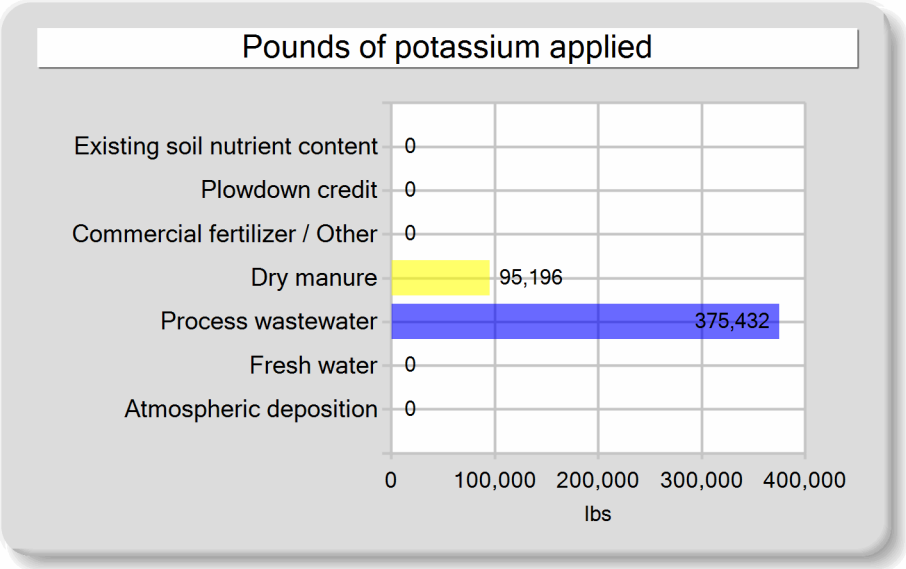
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

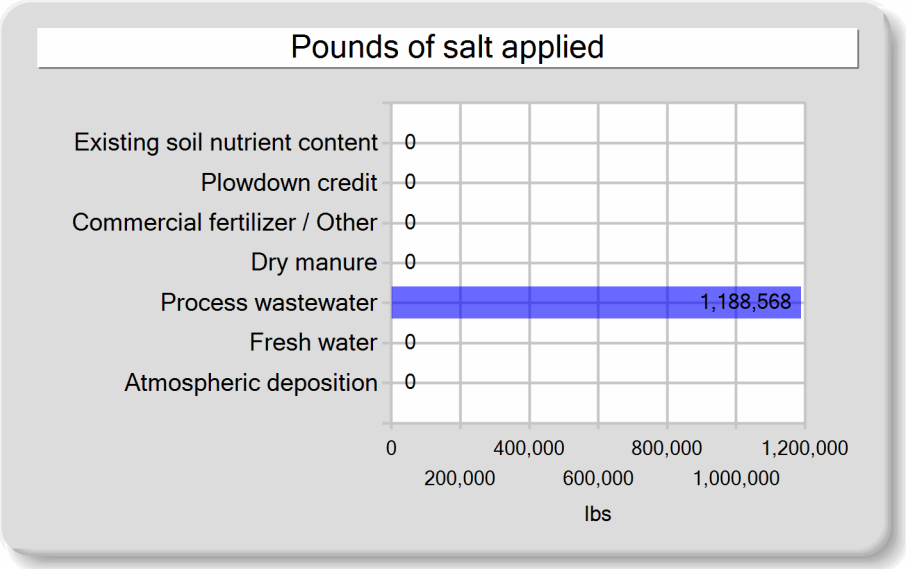


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Frank Goncalves	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 25, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Goncalves Dairy
15728 Arroyo Ave
Dos Palos CA 93620
00-0021361 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L1012
Sampled By: M. Pedroso
Requested By: Frank Goncalves
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - East	12/13/23	14:00	1080	1.08	0.2	ND		8.0
2	DW - West	12/13/23	14:05	1080	1.08	0.2	ND		8.0

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Goncalves Dairy
15728 Arroyo Ave
Dos Palos CA 93620
00-0021361 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L1012
Sampled By: M. Pedroso
Requested By: Frank Goncalves
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	**EC** µmhos/cm	**EC** mmhos/cm	**NO_3-N** mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - East	12/13/23	14:00	**1080**	**1.08**	0.2	0.55	0.05
2	DW - West	12/13/23	14:05	**1080**	**1.08**	0.2	0.55	0.05

Water for Crop Use	Total	Total	Nitrate
Nutrient Management	Salts	Salts	Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.

Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).

Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.

MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)

MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.

SM = Standard Methods for the Examination of Water and Wastewater

EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Goncalves Dairy
15728 Arroyo Ave
Dos Palos, CA 93620

Account# 00-0021361
Account Manager: Ben Nydam
Submitted By: Frank Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:53

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L1012-01	DW - East	Ag Water	M. Pedroso		12/13/2023 14:00
23L1012-02	DW - West	Ag Water	M. Pedroso		12/13/2023 14:05

Default Cooler Temperature on Receipt °C: 1.8
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

Goncalves Dairy
15728 Arroyo Ave
Dos Palos, CA 93620

Account# 00-0021361
Account Manager: Ben Nydam
Submitted By: Frank Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:53

Sample Results

Sample: DW - East
23L1012-01 (Water)

Sampled: 12/13/2023 14:00
Sampled By: M. Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.08**	mmhos/cm	0.01	1		12/19/23 11:56	SM 2510 B		BEL0706
Electrical Conductivity umhos	**1080**	umhos/cm	10.0	1		12/19/23 11:56	SM 2510 B		BEL0706
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 14:00	Field		BEL0653
Nitrate Nitrogen as NO3N	**0.2**	mg/L	0.1	1	10	12/18/23 23:21	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:56	SM 4500-H+	H	BEL0706
pH	**8.0**	units	1.0	1		12/19/23 11:56	SM 4500-H+	H	BEL0706

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DELLAVALLE™
LABORATORY INC

Goncalves Dairy
15728 Arroyo Ave
Dos Palos, CA 93620

Account# 00-0021361
Account Manager: Ben Nydam
Submitted By: Frank Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:53

Sample Results
(Continued)

Sample: DW - West
23L1012-02 (Water)

Sampled: 12/13/2023 14:05
Sampled By: M. Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.08**	mmhos/cm	0.01	1		12/19/23 11:58	SM 2510 B		BEL0706
Electrical Conductivity umhos	**1080**	umhos/cm	10.0	1		12/19/23 11:58	SM 2510 B		BEL0706
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 14:05	Field		BEL0653
Nitrate Nitrogen as NO3N	**0.2**	mg/L	0.1	1	10	12/18/23 23:41	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:58	SM 4500-H+	H	BEL0706
pH	**8.0**	units	1.0	1		12/19/23 11:58	SM 4500-H+	H	BEL0706

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DELLAVALLE™ LABORATORY INC

Goncalves Dairy
15728 Arroyo Ave
Dos Palos, CA 93620

Account# 00-0021361
Account Manager: Ben Nydam
Submitted By: Frank Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:53

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0686										
Blank (BEL0686-BLK1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0686-BLK2)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0686-BLK3)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0686-BS1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		101	90-110		
LCS (BEL0686-BS2)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
Duplicate (BEL0686-DUP1)		**Source: 23L1009-01**			Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L		5.1			0.974	10
Duplicate (BEL0686-DUP2)		**Source: 23L1009-08**			Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L		5.3			0.530	10
Matrix Spike (BEL0686-MS1)		**Source: 23L1009-01**			Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.4		0.1	mg/L	5.000	5.1	106	90-110		
Matrix Spike (BEL0686-MS2)		**Source: 23L1009-08**			Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	10.4		0.1	mg/L	5.000	5.3	102	90-110		
Reference (BEL0686-SRM1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0686-SRM2)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0686-SRM3)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™ LABORATORY INC

Goncalves Dairy
15728 Arroyo Ave
Dos Palos, CA 93620

Account# 00-0021361
Account Manager: Ben Nydam
Submitted By: Frank Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:53

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0706										
Blank (BEL0706-BLK1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.6		1.0	units						
Blank (BEL0706-BLK2)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.9		1.0	units						
Blank (BEL0706-BLK3)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.8		1.0	units						
Duplicate (BEL0706-DUP1)		**Source: 23L1017-01**			Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1.74		0.01	mmhos/cm		1.73			0.403	10
pH	7.7		1.0	units		7.8			0.258	10
Electrical Conductivity umhos	1740		10.0	umhos/cm		1730			0.403	10
Duplicate (BEL0706-DUP2)		**Source: 23L1088-01**			Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	0.52		0.01	mmhos/cm		0.52			0.116	10
Electrical Conductivity umhos	517		10.0	umhos/cm		517			0.116	10
pH	6.7		1.0	units		6.8			0.297	10
Reference (BEL0706-SRM1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	440			umhos/cm	426.0		103	90-110		
Reference (BEL0706-SRM2)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0706-SRM3)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEL0706-SRM4)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0706-SRM5)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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DELLAVALLE™ LABORATORY INC

Goncalves Dairy
15728 Arroyo Ave
Dos Palos, CA 93620

Account# 00-0021361
Account Manager: Ben Nydam
Submitted By: Frank Goncalves

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:53

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0706 (Continued)										
Reference (BEL0706-SRM5)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0706-SRM6)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0706-SRM7)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0706-SRM8)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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12/14/23 14:20 23L1012

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Acct No. / Cons.

Bill To: 21361 | 8

Purchase Order No. _____ Results Needed By _____

Client: **Goncalves Dairy**
Address: 15728 Arroyo Ave
City, State, Zip: Dos Palos, CA 93620
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Frank Goncalves - (209) 617-6324

Ranch _____

Date sampled: 12/13/23

Sampled by: M Rea

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples: 2 No. Bottles: 2

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW - East	Sampled From:	12/13/23	2pm	0	1.8
2. DW West	Sampled From:	12/13/23	205p	0	4.4
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M Rea	CAS Inc	12/13/23 205p	12/14/23 12p
Second	CSM	DLI	12/14/23 12pm	12/14/23 2:15pm
Third				
Fourth	[illegible]	DLI	12/14 14:20	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____ Date ____

Shipping
$ _____ In
$ _____ Out

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1	1								
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/14/23 14:20 23L1012

SEND OUT